DAY 20 OF 30

Semantic models

One agreed meaning for every number.
The layer between data and decisions.

Measures **Relationships** **Star schema**

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THE PROBLEM

Three teams, three reports, three revenue numbers.

1	**Sales**	Counts orders when placed
2	**Finance**	Counts them when paid
3	**Operations**	Counts them when shipped

Everyone is right by their own definition. **The meeting is spent arguing about whose number to use.**

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THE SIMPLE VERSION

A shared dictionary for your data.

A dictionary does not create words. It fixes what each one means, so two people can use the same word and understand each other.

A semantic model fixes what **"revenue", "active customer" and "margin"** mean, once.

✓ **Tables**	The data, tidied for analysis
✓ **Relationships**	How the tables connect
✓ **Measures**	The agreed calculations

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HOW IT WORKS

What a semantic model contains

1. **Fact tables** hold events: sales, visits, readings
2. **Dimension tables** describe them: date, product, region
3. **Relationships** link facts to dimensions
4. **Measures** define each KPI once, in DAX
5. **Reports** read the model, not the raw tables

Define once, use everywhere

A measure written in the model gives the same answer in every report, in Excel and to an AI agent.

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FOR ENGINEERS

What has changed recently

You create it on purpose now

Fabric stopped auto-creating a default semantic model for lakehouses and warehouses. Build the model deliberately.

Storage mode is per table

Import, DirectQuery and Direct Lake can be mixed in one model.

Direct Lake is the default on Fabric data

New models on a warehouse or SQL endpoint use Direct Lake, reading Delta tables without an import.

It is code

Models can be edited as text, versioned in Git and deployed through pipelines.

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THE HONEST PART

Model it or query the tables directly?

BUILD A MODEL

The same KPI appears in many reports

Business users explore the data themselves

An AI agent will answer questions on it

QUERY DIRECTLY

A one-off analysis

Engineers exploring raw data

A single technical report

> **The catch:** a model built on Silver-quality tables spreads bad numbers faster. The model makes a definition consistent. It cannot make it correct.

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FROM REAL WORK

Semantic models on a real project

A people-analytics platform for HR leadership.

1. **Gold tables** in the Lakehouse were shaped for analysis
2. **Semantic models** defined the organisation's KPIs
3. **Row-level security** limited who saw which employees
4. **A GenAI assistant** answered workforce questions in plain language

The assistant was only as good as **the definitions underneath it**.

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SAVE THIS

Semantic models in five lines

1. **One definition** for each business number
2. **Star schema**: facts and dimensions
3. **Measures** hold the logic, not the reports
4. **Created deliberately**, no longer automatic
5. **AI agents** depend on it too

Tomorrow: Direct Lake.
Fast reports without copying the data.

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